

Experimental Results: GRGA Outperforms All Baselines

Category	Method	AliMeeting					AMI Meeting				
		Fact.	Infer.	Temp.	Summ.	Acou.	Fact.	Infer.	Temp.	Summ.	Acou.
Speech LLM	Qwen3-Omni	47.7	63.5	38.3	32.7	11.7	53.6	46.7	37.3	43.6	14.1
	MiMo-Audio	48.0	64.6	38.3	29.4	15.7	53.6	44.1	37.1	41.5	17.9
RAG	BGE-M3 (text)	43.2	29.7	31.6	15.4	18.5	32.6	24.6	23.8	25.4	25.3
	CLASP (audio)	25.4	19.5	18.9	14.8	16.3	24.1	20.3	22.8	11.6	14.8
Ours	GRGA	57.3	66.5	39.5	44.3	39.9	59.5	65.3	38.6	48.5	35.7

Key Findings:

- **+10%** average gain over the best speech LLM on AMI.
- Inferential QA: **65.3%** vs. Text RAG **24.6%**.
- Acoustic QA: **39.9%** vs. next-best **18.5%**.

Why GRGA Wins:

- Graph structure preserves multi-hop meeting dependencies.
- Audio access supports acoustic-aware evidence.
- Reflection suppresses low-evidence answers.